One-Sided Extendability in the Four-Letter Abelian-Square-Free Language

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Abstract

Let $\mathcal{A}_4$ be the language of finite Abelian-square-free words over a four-letter alphabet. We exhibit the explicit word

$$s = \texttt{abacabadc}$$

which cannot be extended by even one letter to the left while remaining Abelian-square-free, but which is a prefix of a right-infinite Abelian-square-free word. Thus

$$s \in re(\mathcal{A}_4) \setminus le(\mathcal{A}_4),$$

and in particular $re(\mathcal{A}_4) \setminus e(\mathcal{A}_4) \neq \varnothing$. The negative half of the proof is elementary. For the positive half we iterate an affine operation built from Keränen's 85-uniform Abelian-square-free endomorphism. Boundary safety is reduced to 99 exact seed geometries, 35 residual states, and 17 recursive transition rows. The proof uses an independently sufficient 178-symbol prefix bound and checks a conservative fixed window of 190 symbols. A finite-certificate verifier and a separate clean-room reconstruction accompany the proof. In the primary sources located, the corresponding one-sided-extension phenomenon was posed as open; no later equivalent resolution was found in the literature searched through 3 September 2026, so the novelty assessment remains provisional.

1 Introduction

An *Abelian square* is a word uv with $|u| = |v| > 0$ in which u and v have the same Parikh vector. Erdős asked whether an infinite word over four letters can avoid Abelian squares. Keränen answered this affirmatively by constructing an 85-uniform morphism g which is itself Abelian-square-free: if w is Abelian-square-free, then so is $g(w)$ [4].

The existence of infinite Abelian-square-free words does not imply that every finite Abelian-square-free word has an infinite continuation. Keränen later studied *unfavourable factors*, finite Abelian-square-free words which cannot occur as proper factors of an infinite Abelian-square-free word. In the 2010 account he noted that such a factor might nevertheless be extendable without bound in one direction, and stated that the existence of such factors remained open at that time [5].

We prove the following concrete asymmetric phenomenon.

Main theorem. The word

$$s = \texttt{abacabadc}$$

is a prefix of a right-infinite Abelian-square-free word, but no letter can be prepended to s without creating an Abelian square.

In the notation for extendable parts of factorial languages used by Shur [7], this gives

$$\boxed{s \in re(\mathcal{A}_4) \setminus le(\mathcal{A}_4)}.$$

Consequently,

$$\boxed{re(\mathcal{A}_4) \setminus e(\mathcal{A}_4) \neq \varnothing}.$$

The proof has deliberately asymmetric difficulty. Left unextendability is checked by four short Abelian squares. Right infinite extendability is obtained from a nested construction

$$C, \quad Cg(C), \quad Cg(C)g^2(C), \quad \dots,$$

where $C = \texttt{abacabadcdb}$. The only nonclassical issue is to prevent new Abelian squares at the repeatedly introduced prefix boundary. We reduce this issue to a finite exact residual certificate.

No claim of minimality is made for the length-9 witness. The historical novelty statement is also kept separate from correctness: in the sources we located the one-sided-extension question was posed as open, but the absence of a later resolution is not itself a proof of novelty.

2 Preliminaries

Let $\Sigma_4 = \{a, b, c, d\}$. For a finite word u, write

$$P(u) = (|u|_a, |u|_b, |u|_c, |u|_d)$$

for its Parikh vector. Words u, v are *Abelian equivalent* if $P(u) = P(v)$. An *Abelian square* is a factor uv with $|u| = |v| > 0$ and $P(u) = P(v)$. Let $\mathcal{A}_4$ denote the factorial language of finite words over Σ_4 containing no Abelian square.

Following Shur [7], the right-, left-, and two-sided extendable parts of a language L consist of words having infinitely many corresponding extensions in L. Equivalently, for our finite alphabet,

$$\begin{aligned}
w \in re(L) &\iff \forall n\ \exists v,\ |v| \geq n,\ wv \in L,\\
w \in le(L) &\iff \forall n\ \exists u,\ |u| \geq n,\ uw \in L,\\
w \in e(L) &\iff \forall n\ \exists u, v,\ |u|, |v| \geq n,\ uwv \in L.
\end{aligned}$$

For a factorial language,

$$e(L) \subseteq re(L) \cap le(L).$$

When a nested infinite right extension is explicitly constructed, membership in $re(L)$ is immediate. Conversely, for a factorial language over a finite alphabet, arbitrarily long compatible finite right extensions yield a one-sided infinite extension by the usual finitely branching tree argument: factoriality makes the continuation tree prefix-closed, finite alphabet makes it finitely branching, and König's lemma supplies an infinite branch.

3 A word which is dead on the left

Set

$$s = \texttt{abacabadc}.$$

A direct check shows that $s \in \mathcal{A}_4$.

Lemma 3.1. *No letter in Σ_4 can be prepended to s while preserving Abelian-square-freeness.*

Proof. The following Abelian squares occur in the four possible extensions.

prepended letter	half-period	forbidden factor	Parikh vector
a	1	$a \mid a$	$(1,0,0,0)$
b	2	$ba \mid ba$	$(1,1,0,0)$
c	4	$caba \mid caba$	$(2,1,1,0)$
d	5	$dabac \mid abadc$	$(2,1,1,1)$

Thus no length-1 left extension belongs to $\mathcal{A}_4$, and hence $s \notin le(\mathcal{A}_4)$. □

4 Keränen's morphism and the right-infinite construction

We use Keränen's 85-uniform Abelian-square-free endomorphism g [4]. Its four images are given in Appendix A. The classical property used here is

$$w \in \mathcal{A}_4 \quad \Longrightarrow \quad g(w) \in \mathcal{A}_4. \tag{4.1}$$

Let

$$C = \texttt{abacabadcdb}$$

and define $F(V) = C\,g(V)$. Starting from $W_0 = C$, put

$$W_{n+1} = F(W_n) = C\,g(W_n).$$

Since g is a morphism,

$$W_n = C\,g(C)\,g^2(C)\cdots g^n(C), \tag{4.2}$$

so W_n is a prefix of W_{n+1}. Hence the limit

$$W_\infty = C\,g(C)\,g^2(C)\cdots \tag{4.3}$$

is well-defined. The only possible new squares under F are those meeting the prefix C, because factors wholly contained in $g(V)$ are covered by (4.1).

5 Residual states

We use row Parikh vectors. Let M be the incidence matrix of g: the row indexed by x is $P(g(x))$. Directly from the four images,

$$M = \begin{pmatrix} 19 & 21 & 27 & 18 \\ 18 & 19 & 21 & 27 \\ 27 & 18 & 19 & 21 \\ 21 & 27 & 18 & 19 \end{pmatrix}, \qquad \det M = 43435 \neq 0. \tag{5.1}$$

Thus $qM = r$ has at most one rational solution q, and integrality can be checked exactly.

Definition 5.1 (Residual occurrence)**.** For $q \in \mathbb{Z}^4$ and $x, y \in \Sigma_4$, a residual state $R(q, x, y)$ occurs in a word V if

$$V = A\,x\,B\,y\,D \tag{5.2}$$

for some words A, B, D, with the displayed occurrences of x and y distinct and in this order, and

$$P(A) - P(B) = q. \tag{5.3}$$

5.1 States forced by a large crossing square

Suppose an Abelian square in $F(V) = Cg(V)$ is not contained wholly in $g(V)$. Its start is at a position $i \in \{0, \ldots, 10\}$ of C. Let its half-period be $K \geq 85$. We may write the relevant part of the source word as

$$V = A\,x\,B\,y\,D,$$

where the midpoint lies at offset $r \in \{0, \ldots, 84\}$ of $g(x)$, and the end lies at offset $t \in \{0, \ldots, 84\}$ of $g(y)$. An end exactly at the final boundary of an image may additionally be represented by $t = 85$; the exact audit produces no additional integral state from this case.

Equality of the two half-Parikh vectors gives

$$qM = P(g(x)[r:]) + P(g(y)[:t]) - P(C[i:]) - P(g(x)[:r]), \tag{5.4}$$

where $q = P(A) - P(B)$.

We define Q to be the set of all integral triples (q, x, y) obtained from (5.4) as the finite parameters vary. Exact integer inversion gives 99 valid parameter rows collapsing to

$$|Q| = 35 \tag{5.5}$$

unique residual states. Their complete list and the 99 geometric rows are supplied as machine-readable supplementary data.

Lemma 5.2 (Large crossing-square reduction)**.** *If V is Abelian-square-free and $Cg(V)$ contains an Abelian square meeting C with half-period at least 85, then V contains a residual state from Q.*

Proof. Choose i, x, r, y, t from the actual square and set $q = P(A) - P(B)$. Equation (5.4) holds. Since $q \in \mathbb{Z}^4$, this actual configuration is among the exact integral cases used to define Q. Hence $R(q, x, y)$ occurs in V. □

6 Recursive closure of the residual states

Suppose $R(q, x, y) \in Q$ occurs in $Cg(V)$ and the two marked target letters lie in distinct image blocks $g(h)$ and $g(k)$, at offsets r and t. Write the corresponding source factorization as

$$V = A\,h\,B\,k\,D,$$

and set $q' = P(A) - P(B)$. Expanding the target prefix and gap gives

$$q'M = q - P(C) - P(g(h)[:r]) + P(g(h)[r+1:]) + P(g(k)[:t]). \tag{6.1}$$

The omission of the marked target letter itself explains the suffix $g(h)[r+1:]$.

6.1 Worked transition: from a certificate row back to word geometry

Consider the genuine target state

$$R((0,1,0,-1), d, b)$$

with offsets $r = 68$ in $g(h)$ and $t = 63$ in $g(k)$, where $h = k = b$. The source factorization is $V = A\,b\,B\,b\,D$. Equation (6.1) uses

$$\begin{aligned} q &= (0,1,0,-1), & P(C) &= (4,3,2,2),\\ P(g(b)[:68]) &= (13,16,18,21), & P(g(b)[69:]) &= (5,3,3,5),\\ P(g(b)[:63]) &= (12,15,17,19). \end{aligned}$$

Consequently,

$$\begin{aligned} q'M &= (0,1,0,-1) - (4,3,2,2) - (13,16,18,21) \\ &\qquad + (5,3,3,5) + (12,15,17,19) \\ &= (0,0,0,0). \end{aligned}$$

Since $\det M = 43435 \neq 0$, this gives $q' = (0,0,0,0)$. Thus the row desubstitutes to

$$R((0,0,0,0), b, b)$$

in V. The row is a machine-certified exact witness for the transition equation; the displayed arithmetic is paper-and-pencil verifiable.

Put $m = |A|$. The first source mark is at position m, while the first target mark is at

$$j = |C| + 85m + r = 11 + 85m + 68 = 79 + 85m.$$

Hence

$$j - m = 79 + 84m \geq 79. \tag{6.2}$$

This gives the meaning of one transition row and the minimum exact descent margin reported by the clean-room reconstruction.

Exact inversion of (6.1), over every state in Q and every compatible choice of h, r, k, t, produces exactly 17 integral parameter rows. Every source triple (q', h, k) belongs again to Q.

Lemma 6.1 (Recursive residual closure)**.** *If a state from Q occurs in $Cg(V)$ with its marked letters in distinct image blocks beyond C, then a state from Q occurs in V.*

Proof. For an actual occurrence, the source vector $q' = P(A) - P(B)$ is integral and satisfies (6.1). The exact finite inversion therefore includes the actual alignment; every integral source state produced by that enumeration lies in Q. □

7 Base cases and strict descent

Lemma 7.1 (Fixed base window)**.** *Every crossing Abelian square of half-period $K < 85$, and every residual occurrence from Q which does not fall under the recursive residual lemma, is contained in the first 178 letters of $Cg(V)$, provided that V begins with C. The finite certificate checks the larger fixed prefix of length 190.*

Proof. A crossing square starts at a zero-based position $i \leq 10$. If $K < 85$, its exclusive end is at most $10 + 2 \cdot 84 = 178$.

Now consider a residual occurrence

$$Cg(V) = A\,x\,B\,y\,D$$

with marked positions $j < k$. From $q = P(A) - P(B)$, taking coordinate sums gives

$$\sum q = j - (k - j - 1) = 2j + 1 - k,$$

so

$$k = 2j + 1 - \sum q. \tag{7.1}$$

The 35-state certificate satisfies

$$\sum q \in \{-1, 0, 1\} \tag{7.2}$$

for every q in Q. If the first marked letter lies in C, then $j \leq 10$, so $k \leq 22$; these cases lie in a prefix of length 23. If the first mark lies beyond C but both marks lie in the same 85-letter image block, put $d = k - j \leq 84$. Equation (7.1) gives $j = d + \sum q - 1 \leq 84$, and then $k \leq 168$. These cases lie in a prefix of length 169. Thus the independently sufficient prefix lengths are 178 for short crossing squares, 23 for marks in C, and 169 for same-block residuals. Their maximum is 178. The certificate intentionally checks the conservative 190-symbol prefix. No minimality claim is made for the 190-symbol window. $\square$

Because the invariant below requires V to begin with C, the first 190 letters of $Cg(V)$ equal the first 190 letters of $W_1 = Cg(C)$. The exact verifier checks that this fixed word is Abelian-square-free and contains no state from Q.

The recursive step is well-founded without an approximate contraction estimate. If the first marked position in a recursive target occurrence is $j \geq 11$, its containing source letter occurs at position

$$a = \left\lfloor \frac{j - 11}{85} \right\rfloor$$

of V. Therefore $a < j$. Recursive desubstitution strictly decreases a nonnegative integer and cannot continue indefinitely. The clean-room reconstruction evaluates all 17 rows and finds a minimum exact margin of 79, attained by the worked row (6.2); it finds no nondecreasing realized transition.

If positions are forgotten, the state graph has one cyclic strongly connected component,

$$\{R((0, 0, -1, 0), c, b), R((0, 1, 0, -1), d, b)\}.$$

This state-only cycle does not threaten termination. A graph edge can be traversed by an actual occurrence only together with its image-block position, and every such realized transition decreases the nonnegative occurrence-position measure by at least 79. Graph-theoretic cycling therefore does not imply occurrence-level cycling.

8 The invariant and the infinite word

Define

$$\mathcal{C}^* = \{V : \ C \text{ is a prefix of } V, \ V \in \mathcal{A}_4, \text{ and } V \text{ contains no state from } Q\}. \tag{8.1}$$

The fixed-prefix condition is essential: it makes all short cases reduce to one fixed base window.

Proposition 8.1 (Invariance). *If $V \in \mathcal{C}^*$, then $Cg(V) \in \mathcal{C}^*$.*

Proof. The word $Cg(V)$ begins with C. Since V is Abelian-square-free, (4.1) implies that $g(V)$ is Abelian-square-free. Therefore any Abelian square in $Cg(V)$ must meet C. If its half-period is below 85, the fixed-base lemma places it in the 190-letter base window, which the certificate verifies to be Abelian-square-free. If its half-period is at least 85, the large crossing-square lemma would force a state from Q in V, contrary to $V \in \mathcal{C}^*$.

It remains to exclude states from Q in $Cg(V)$. A nonrecursive occurrence lies in the fixed base window and is excluded by the certificate. A recursive occurrence would, by recursive residual closure, force a state from Q in V, again a contradiction. Thus $Cg(V) \in \mathcal{C}^*$. $\square$

Proposition 8.2. *The word C belongs to $\mathcal{C}^*$.*

Proof. The exact finite verifier checks directly that C is Abelian-square-free and contains none of the 35 states from Q. $\square$

Corollary 8.3. *Every W_n is Abelian-square-free, and the limit W_∞ is a right-infinite Abelian-square-free word.*

Proof. The preceding propositions imply $W_n \in \mathcal{C}^* \subseteq \mathcal{A}_4$ for all n. Any finite factor of the nested limit W_∞ occurs in some W_n, so the limit contains no Abelian square. □

9 Main result

Theorem 9.1. *For $s = \texttt{abacabadc}$,*

$$\boxed{s \in re(\mathcal{A}_4) \setminus le(\mathcal{A}_4)}.$$

Proof. The word s is a prefix of C, hence a prefix of the right-infinite Abelian-square-free word W_∞. Thus $s \in re(\mathcal{A}_4)$. The left-death lemma gives $s \notin le(\mathcal{A}_4)$. □

Corollary 9.2.

$$re(\mathcal{A}_4) \setminus e(\mathcal{A}_4) \neq \varnothing.$$

Proof. Since $e(\mathcal{A}_4) \subseteq le(\mathcal{A}_4)$, the theorem gives the claim. □

10 Computer-assisted part and reproducibility

The proof uses one classical external input, Keränen's theorem (4.1), and one finite exact computation specific to the boundary C. The author-side script `verify_p7_main_theorem_v4.py` recomputes from the displayed formulas:

1. the four immediate left-death witnesses;
2. the incidence matrix M and exact integer inversion;
3. the 99 large-square seed alignments;
4. the 35 unique residual states Q;
5. the 17 integral recursive transition rows and closure back into Q;
6. the bound $\sum q \in \{-1, 0, 1\}$;
7. absence of Abelian squares and residual states in the fixed 190-letter base window;
8. membership of C in the invariant.

It also performs direct regression checks on W_1 and on boundary-crossing squares in W_2. These finite long-word checks are not used as extrapolation in the all-depth proof.

A representative successful run reports:

```
P7 V2 FINITE CERTIFICATE: PASS
left death: 4/4
seed parameter rows: 99
residual states: 35
recursive transition rows: 17
base window: 190 symbols - no Abelian-square or residual violation found
W1: 946 ASF
W2: 80421 no crossing square
FINITE CERTIFICATE RECONSTRUCTED AND VERIFIED
```

This output concerns the finite certificate only. The script does not independently prove Keränen's external morphism theorem, the prose proof of universal invariant closure, or historical novelty.

A separate clean-room implementation, `P7_CODEX_CLEANROOM_VERIFIER.py`, was written without importing or parsing the author verifier or the submitted CSV certificates. It independently reconstructed 99 seed geometry rows, 35 residual states, and 17 recursive transition rows, then found exact set equality with the supplied data. It also reported minimum descent margin 79, the one two-state state-only SCC displayed above, no nondecreasing realized transition, and independently sufficient base-prefix length 178. This is an independent computational reconstruction, not human peer review.

Obligation	**Source**
g_{85} preserves ASF	Keränen 1992
left death of s	direct finite proof
seed completeness	finite exact reconstruction
residual closure	finite exact reconstruction
short/base cases	direct finite check
infinite limit	mathematical induction

The exact state and transition tables are supplied as machine-readable supplementary files.

11 Relation to previous work

Keränen's 1992 construction established an Abelian-square-free endomorphism on four letters and therefore infinite four-letter Abelian-square-free words [4]. Carpi later showed, among other results, exponential growth in the number of four-letter Abelian-square-free words [2].

The finite extension problem has a distinct literature. Cummings and Mays constructed finite maximal Abelian-square-free words [3]; Korn gave short maximal constructions and bounds [6], later improved by Bullock [1]. That literature also defines and studies one-sided maximality, but it does not provide the combination established here: a word dead on the left together with an infinite Abelian-square-free continuation on the right.

Keränen's later work on unfavourable factors is closer to the present question. In the 2010 primary source located for this audit, he explicitly observed that an unfavourable factor might still be extendable without bound in one direction and stated that the existence of such factors remained open at that time [5]. The theorem gives a positive answer, in a stronger form, to the one-sided-extension phenomenon posed by Keränen: the witness has a right-infinite continuation while admitting no legal one-letter extension on the left.

Shur's framework makes the set-theoretic consequence transparent: despite general results comparing growth rates of extendable parts of factorial languages, the actual sets $re(\mathcal{A}_4)$ and $e(\mathcal{A}_4)$ need not coincide [7].

The literature audit used primary sources together with forward-citation and alternate-term searches through 3 September 2026. It found no later result equivalent to the theorem. Direct exhaustive access to all closed citation indexes, including Scopus and Web of Science, was not available. The resulting status is *No prior resolution found - novelty provisional*, not a proof of novelty. An independent domain-expert or librarian citation review is recommended before final submission language is frozen.

12 Discussion

The witness is small and its obstruction is completely local on the left, whereas the proof of infinite survival on the right is global and computer-assisted. Natural further questions include the minimum possible witness length, structural classification of left-unextendable/right-infinite words, and whether a shorter human proof of the right-infinite construction exists. No minimality or uniqueness claim is made for s, C, or the construction.

A The 85-uniform morphism

$g(a) =$ `abcacdcbcdcadcdbdabacabadbabcbdbcbacbcdcacbabdabacadcbcdcacdbcbacbcdcacdcbdcdadbdcbca`

$g(b) =$ `bcdbdadcdadbadacabcbdbcbacbcdcacdcbdcdadbdcbcabcbdbadcdadbdacdcbdcdadbdadcadabacadcdb`

$g(c) =$ `cdacabadabacbabdbcdcacdcbdcdadbdadcadabacadcdbcdcacbadabacabdadcadabacabadbabcbdbadac`

$g(d) =$ `dabdbcbabcbdcbcacdadbdadcadabacabadbabcbdbadacdadbdcbabcbdbcabadbabcbdbcbacbcdcacbabd`

B Finite certificate files and provenance

The supplementary proof data are `P7_V2_RESIDUAL_STATES.csv` (35 states), `P7_V2_SEED_ROWS.csv` (99 seed rows), `P7_V2_RECURSIVE_TRANSITIONS.csv` (17 recursive rows), `verify_p7_main_theorem_v4.py` (author-side finite verifier), `P7_CODEX_CLEANROOM_VERIFIER.py` (independent clean-room reconstruction), and `P7_CODEX_HIGH_CERTIFICATE_RECONSTRUCTION.md` (reconstruction equations, bounds, outputs, and digests).

The historical v0.1 proof package is rejected and is not evidence for this theorem. Its prefix-free invariant admitted $V = b$, which is Abelian-square-free, but $Cg(b)$ contains the boundary Abelian square `bb` beginning at zero-based position 10. The v0.1 residual expression also had defective word-level semantics. The corrected invariant requires C to be a prefix of V. This excludes $V = b$ and makes the finite base prefix of $Cg(V)$ universal across the invariant class.

The canonical release contains exactly one manuscript PDF. Its SHA-256 is recorded in `SHA256SUMS.txt`; any PDF outside that release directory is noncanonical. The authorship line remains the only unresolved metadata field and must be replaced before submission.

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